

IP Internet protocol location
 MAC Media access control Identification of data } network
 Getmac

MAC

- Media access control
- It is a physical address
- It is in hexa decimal format
- It contains 12 hexa decimal
- Each hexa contains 4 bit
- 12*4 = 48 bit length

MAC

IC-C1-DE-33-8D-14

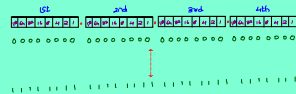
48 bit
 24 bit 24 bit
 Vendor ID User ID
 S Suseen

IP

- Internet protocol
- It is a logical address
- It is in dotted decimal format
- It contains 4 octet
- Each octet contains 8 bit
- 8*4 = 32 bit length
- IP = network + host

configuration

IPv4 Structure



Binary combination
 32 bit

0.0.0.0 Starting IP
 255.255.255.255 Ending IP
 Class A B C D E

CLASS	RANGE	PORTION
A	0 - 127	N.H.H.H
B	128 - 191	N.N.H.H
C	192 - 223	N.N.N.H
D	224 - 239	Multicast
E	240 - 255	Broadcast (R/D)

Included IP

Type of Communication

Unicast one to one
 Multicast one to group
 Broadcast one to all

12.0.0.86 - A - N.H.H.H

12.10.234.254 - B - N.N.H.H

192.10.192.51 - C - N.N.N.H

50.0.0.10
 N.H.H.H

Network - Street no
 Host - Door no

0 0 0 0
 192.192.192.192
 N.N.N.H

How many n/w bit?

24

How many host octet?

1

How many host bit?
 8
 How many n/w octet?
 3

fail?
 cross?
 G12
 Thank
 Good

Suction
 G11
 Thanku
 Note
 Good

2 bit Binary

2	1
0	0
0	1
1	0
1	1

3 bit binary

4	2	1
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

5 bit binary

16	8	4	2	1	
0	0	0	0	0	0
0	0	0	0	1	1
0	0	0	1	0	2
0	0	0	1	1	3
0	0	1	0	0	4
0	0	1	0	1	5
					6
					7
					8